# Sao Aphisith Sithisack

Norman, OK | (572) 236-9592 | sao.aphisith.sithisack-1@ou.edu
saosith.github.io | linkedin.com/in/saoaphisithsithisack | github.com/saosith

## EDUCATION

**University of Oklahoma** **Norman, OK**
B.S. in Electrical Engineering (Honors) | GPA: 3.88 *Aug 2024 – May 2028*
**Honors:** Davis UWC Scholar (full tuition, 4 yrs) | Robert Vinton Clapp Scholarship
**Relevant Coursework:** Digital Design, Digital Signal Processing, Introduction to Electronics, Microcontrollers, Circuits Laboratory, Electromagnetic Fields, Differential Equations, Linear Algebra

## EXPERIENCE

**Pulsed Laser Template Engineering (PLATEN) Lab** **Jan 2026 – Present**
*Undergraduate Researcher, Semiconductor Nanofabrication* *Norman, OK*
- Fabricate high-aspect-ratio micro- and nanostructures in a cleanroom using photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Independently own the PLD step end to end, growing LAO films on STO substrates and monitoring growth with RHEED.
- Optimize anisotropic plasma etch parameters (gas chemistry, RF power, chamber pressure) and mask selectivity to reach vertical sidewalls and controlled feature depth.

**Sooner Competitive Robotics** **Jan 2026 – Present**
*Electrical Systems Engineer, Power Distribution* *Norman, OK*
- Design a **4-layer** power, CAN, and E-Stop distribution PCB in **Altium** for the IGVC 2027 robot, routing 12 V at up to **10 A** to **7 board ports**, each with INA220 current sensing (layout in progress).
- Replaced the team-standard Schottky with P-MOSFET reverse-polarity protection, cutting calculated loss at 8 A from **3.6 W to 0.7 W**.
- Specified the 3.3 V synchronous buck converter and the 8 MHz crystal clock for the 125 kbps CAN bus, recording each design decision for team and captain review.

**Student Learning Center** **Aug 2025 – Present**
*Peer Learning Assistant | 20 hrs/week* *Norman, OK*
- Mentor **250+ students per semester** in Physics I–II, Calculus I–III, Differential Equations, Linear Algebra, and General Chemistry I through structured problem-solving.

**Cloud Microphysics Group, First Year Research Experience** **Jan 2025 – May 2025**
*Undergraduate Researcher, Data Analysis* *Norman, OK*
- Processed NSF/NCAR C-130 aircraft data from the CAESAR campaign in **Python**, comparing cloud water content, particle size, and condensation-nuclei concentration across probes versus altitude; presented the results as a research poster.

## PROJECTS

**TI MSPM0 USB-C Microcontroller Board** **Sep 2026 – Present**
*KiCad | 2-layer PCB | write-up* *Personal Project*
- Designed the schematic for a USB-C powered **Texas Instruments MSPM0G3507** board: 3.3 V LDO, CH340E USB-to-UART bridge, USBLC6 ESD protection, 8 MHz crystal, and SWD debug; routing the 2-layer layout.
- Set design targets from USB 2.0 limits and datasheets: $\leq$100 mA load, $\leq$0.2 W LDO loss, ADC reference noise $<$1 LSB.

**Discrete Analog Function Generator** **Oct 2026 – Present**
*LTspice | in progress, not yet built* *Personal Project*
- Simulated the oscillator core in LTspice, running **Monte Carlo** on resistor tolerance to justify 1% parts: simulated frequency spread narrowed from about $-$12%/+9% at 5% resistors to about $\pm$2%.

## LEADERSHIP

**United World College Lao PDR National Committee** **Jul 2024 – Present**
*Co-President & Alumni Coordinator* *Remote / Laos*
- Co-lead a **20+ volunteer** committee running Laos' UWC scholarship cycle, managing **150+ applicants** per cycle down to 3–4 placements, and led fundraising exceeding **$2,500**.

**Association of Southeast Asian Nations at OU** **Nov 2024 – May 2026**
*Sponsorship & Volunteering Chair* *Norman, OK*
- Secured $1,000+ in external funding through direct outreach to 20+ donors and organized cultural events.

**Activities:** IEEE Student Branch at OU, Event Committee | Sooner Racing Team, Electrical (Sep 2026 – Present)

## TECHNICAL SKILLS

**PCB & Hardware:** Altium Designer (4-layer PCB), KiCad (schematic capture), power protection and distribution, USB-C, UART, CAN, $I^2C$, SWD, LTspice (Monte Carlo)
**Fabrication & Lab:** Photolithography, anisotropic RIE, pulsed laser deposition, RHEED, cleanroom processing, oscilloscope, DMM, function generator, bench supply
**Programming:** C, C++, Python, MATLAB, LabVIEW, Linux, Git